

Simplex Binarizations and Rank: A Short Solution Note

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Abstract

Aprile, Conforti, and Di Summa introduced a rank parameter for natural binarizations and asked whether the unary binarization is optimal among binarizations whose polytope is isomorphic to a simplex. This note gives a self-contained resolution of the literal single-gap formulation and a sharper statement for the multi-gap rank function. If $A = \{\alpha_1 < \dots < \alpha_\ell\} \subseteq \{0, \dots, d-1\}$ is a set of integer gaps, then among all natural binarizations of $x \in \{0, \dots, d\}$ whose polytope is a d -simplex, the minimum possible rank for the gap set A is

$$\lceil \log_2(\ell + 1) \rceil.$$

Consequently the unary binarization is optimal for every single gap, but it is not pointwise optimal for the multi-gap rank once $\ell \geq 3$. The unary binarization is nevertheless Pareto-optimal for the family of all single-gap ranks: matching unary's value for every individual gap forces the unary code up to permutation and complementation of coordinates.

1 Introduction

Let x be an integer variable with range $\{0, 1, \dots, d\}$. A binarization represents the integrality of x through auxiliary variables $y \in [0, 1]^d$ that are intended to be binary. Aprile, Conforti, and Di Summa introduced a notion of *natural* binarization and a rank parameter $\text{rk}_B(\alpha_1, \dots, \alpha_\ell)$ measuring how many binary variables must be sequentially convexified in order to eliminate vertices whose x -coordinate lies across one of the listed unit gaps; they also gave a set-covering characterization of this rank [1]. Their first open question asks whether the unary binarization $B_U(d)$ is optimal among simplex-isomorphic binarizations [1]. The same question is recorded in the Open Problems in Operations Research collection [2].

This note isolates the finite combinatorial core of the question. For simplex-isomorphic natural binarizations of $x \in \{0, \dots, d\}$, the vertices are exactly an ordered list of $d + 1$ distinct binary codewords

$$b^0, b^1, \dots, b^d \in \{0, 1\}^d,$$

with one codeword for each value of x . For a fixed set of gaps, the rank asks for the smallest set of coordinates whose restrictions distinguish all codewords lying in different blocks cut out by those gaps. This immediately turns the problem into an elementary coding question.

The main result is that, for ℓ distinct gaps, the exact best possible rank is $\lceil \log_2(\ell + 1) \rceil$. The literal single-gap formulation is therefore answered affirmatively: unary has rank 1, and no genuine gap can have rank less than 1. The stronger multi-gap pointwise statement is false: unary has multi-gap rank ℓ , whereas a suitable simplex code attains only $\lceil \log_2(\ell + 1) \rceil$.

2 From simplex binarizations to ordered binary codes

We use the following definition. A polytope $B \subseteq \mathbb{R} \times [0, 1]^d$ is a binarization of $x \in \{0, \dots, d\}$ if

$$\pi_x(B \cap (\mathbb{R} \times \{0, 1\}^d)) = \{0, 1, \dots, d\}.$$

It is natural if every vertex of B has integral x -coordinate. In this note, *simplex binarization* means a natural binarization whose polytope is a d -simplex. Let $\mathcal{S}_d$ denote the class of all such binarizations.

Lemma 2.1 (code representation). *Let $B \in \mathcal{S}_d$. Then its vertices can be written uniquely as*

$$(0, b^0), (1, b^1), \dots, (d, b^d),$$

where $b^0, \dots, b^d \in \{0, 1\}^d$ are distinct. Conversely, if $b^0, \dots, b^d \in \{0, 1\}^d$ are distinct and the points $(0, b^0), \dots, (d, b^d)$ are affinely independent, then

$$B = \text{conv}\{(i, b^i) : i = 0, \dots, d\}$$

is a simplex binarization of $x \in \{0, \dots, d\}$.

Proof. Let $(x, b) \in B$ with $b \in \{0, 1\}^d$, and write it as a convex combination of vertices:

$$(x, b) = \sum_r \lambda_r (x^r, y^r), \quad \lambda_r \geq 0, \quad \sum_r \lambda_r = 1.$$

For any coordinate j , if $b_j = 0$, then every vertex used with positive coefficient must have $y_j^r = 0$, because all $y_j^r \in [0, 1]$. Similarly, if $b_j = 1$, every used vertex must have $y_j^r = 1$. Hence every vertex used with positive coefficient has the same y -vector, namely b .

If two different vertices with this same binary y -vector were used, then the segment between them would contain points with the same binary y -vector and nonintegral x -coordinate. This would contradict the binarization identity $\pi_x(B \cap (\mathbb{R} \times \{0, 1\}^d)) = \{0, \dots, d\}$. Therefore every point of B with binary y is itself a vertex.

Since B is a d -simplex, it has exactly $d + 1$ vertices. Naturalness says that their x -coordinates are integral, and the binarization property requires a binary feasible point above each $i \in \{0, \dots, d\}$. By the previous paragraph these binary feasible points are vertices, so there is exactly one vertex above each integer i , and it has the form (i, b^i) with $b^i \in \{0, 1\}^d$. The codewords b^i are distinct; otherwise the segment between (i, b^i) and (j, b^j) , $i \neq j$, would again give nonintegral x -values with binary y .

Conversely, suppose that $(0, b^0), \dots, (d, b^d)$ are affinely independent with distinct binary codewords. Their convex hull is a d -simplex, all of whose vertices have integral x -coordinate. If a convex combination of these vertices has binary y , the same coordinatewise argument shows that all used codewords must equal that binary vector. Since the codewords are distinct, only one vertex is used. Thus the only points of B with binary y are the displayed vertices, and their x -coordinates are precisely $0, \dots, d$. Hence $B \in \mathcal{S}_d$. $\square$

Let $A = \{\alpha_1 < \dots < \alpha_\ell\} \subseteq \{0, \dots, d-1\}$. These gaps partition $\{0, \dots, d\}$ into $\ell + 1$ nonempty consecutive blocks:

$$\begin{aligned} I_0 &= \{0, \dots, \alpha_1\}, \\ I_1 &= \{\alpha_1 + 1, \dots, \alpha_2\}, \\ &\vdots \\ I_\ell &= \{\alpha_\ell + 1, \dots, d\}. \end{aligned}$$

For vertices (i, b^i) and (j, b^j) , define the indicator set

$$D_{ij} = \{q \in [d] : b_q^i \neq b_q^j\}.$$

Because a simplex has complete graph skeleton, every pair of vertices is an edge. An edge is separated by the gap set A exactly when its two x -indices lie in different blocks.

We use the set-covering characterization of rank from [1]: for a simplex binarization represented by a code $b^0, \dots, b^d$,

$$\text{rk}_B(A) = \min\{|S| : S \subseteq [d] \text{ and } S \cap D_{ij} \neq \emptyset \text{ whenever } i, j \text{ lie in different blocks}\}. \quad (1)$$

Equivalently, S is feasible in (1) if and only if

$$b^i|_S \neq b^j|_S$$

whenever i and j lie in different blocks.

3 The exact optimum for a fixed gap set

Theorem 3.1 (exact optimum). *Let $A \subseteq \{0, \dots, d-1\}$ be a set of ℓ distinct gaps. Then*

$$\min_{B \in \mathcal{S}_d} \text{rk}_B(A) = \lceil \log_2(\ell + 1) \rceil.$$

For $A = \emptyset$, both sides are interpreted as 0.

Proof. The empty case is immediate, so assume $\ell \geq 1$. Let $m = \lceil \log_2(\ell + 1) \rceil$.

First we prove the lower bound. Let $B \in \mathcal{S}_d$, and let $S \subseteq [d]$ be feasible in the set-cover formulation (1). Choose one index $i_r \in I_r$ from each block I_r , $r = 0, \dots, \ell$. If $r \neq s$, then i_r and i_s are in different blocks, so feasibility of S implies

$$b^{i_r}|_S \neq b^{i_s}|_S.$$

Thus the $\ell + 1$ selected blocks give $\ell + 1$ distinct binary strings of length $|S|$. Hence

$$2^{|S|} \geq \ell + 1,$$

and therefore $|S| \geq m$. This proves $\text{rk}_B(A) \geq m$ for every $B \in \mathcal{S}_d$.

It remains to construct a simplex binarization with rank at most m . Choose $\ell + 1$ distinct labels

$$\lambda_0, \dots, \lambda_\ell \in \{0, 1\}^m$$

so that $\lambda_0 = 0$ and the standard unit vectors $e_1, \dots, e_m$ occur among the labels. This is possible because $\ell + 1 \geq m + 1$ and $2^m \geq \ell + 1$.

We now define codewords $b^0, \dots, b^d \in \{0, 1\}^d$. For every $i \in I_r$, set the first m coordinates of b^i equal to λ_r . Choose one base index $p_0 \in I_0$, and for each $j = 1, \dots, m$, choose one index p_j from the block whose label is e_j . For these $m + 1$ special indices, put zeroes in all remaining coordinates $m + 1, \dots, d$. There are exactly $d - m$ other indices. Give each of them a private coordinate among $m + 1, \dots, d$, putting a 1 in that private coordinate and 0's in all other remaining coordinates.

The resulting codewords are affinely independent. Indeed, $b^{p_0} = 0$. The vectors $b^{p_j} - b^{p_0}$, $j = 1, \dots, m$, are the first m standard basis vectors. Every nonspecial codeword has a private standard basis coordinate among the last $d - m$ coordinates, so any linear dependence among $\{b^i - b^{p_0} : i \neq p_0\}$ has zero coefficient on all nonspecial codewords, and then zero coefficient on each b^{p_j} as well. Hence the codewords are affinely independent in $\mathbb{R}^d$, and therefore the points (i, b^i) are affinely independent in $\mathbb{R} \times \mathbb{R}^d$.

By Lemma 2.1,

$$B = \text{conv}\{(i, b^i) : i = 0, \dots, d\}$$

is a simplex binarization. Selecting the first m coordinates separates all blocks because distinct blocks have distinct labels λ_r . Therefore $\text{rk}_B(A) \leq m$. Together with the lower bound, this gives

$$\min_{B \in \mathcal{S}_d} \text{rk}_B(A) = m = \lceil \log_2(\ell + 1) \rceil.$$

□

4 Consequences for unary binarization

The unary binarization $B_U(d)$ has vertices

$$(i, u^i), \quad u^i = (\underbrace{1, \dots, 1}_i, 0, \dots, 0) \in \{0, 1\}^d, \quad i = 0, \dots, d.$$

Lemma 4.1 (unary rank). *For every gap set $A \subseteq \{0, \dots, d - 1\}$,*

$$\text{rk}_{B_U(d)}(A) = |A|.$$

Proof. Let $A = \{\alpha_1, \dots, \alpha_\ell\}$. The edge between consecutive vertices α and $\alpha + 1$ has indicator set

$$D_{\alpha, \alpha+1} = \{\alpha + 1\}.$$

Thus any feasible hitting set for A must contain the coordinate $\alpha + 1$ for every $\alpha \in A$, so $\text{rk}_{B_U(d)}(A) \geq |A|$. These same coordinates also suffice: if two unary codewords lie on different sides of a gap $\alpha \in A$, then they differ in coordinate $\alpha + 1$. Hence $\text{rk}_{B_U(d)}(A) = |A|$. □

Corollary 4.2 (single gaps). *For every simplex binarization $B \in \mathcal{S}_d$ and every single gap $\alpha \in \{0, \dots, d - 1\}$,*

$$\text{rk}_B(\{\alpha\}) \geq \text{rk}_{B_U(d)}(\{\alpha\}) = 1.$$

Thus the unary binarization is optimal for the literal single-gap formulation.

Proof. Apply Theorem 3.1 with $\ell = 1$, or equivalently observe that at least one coordinate is needed to separate the adjacent vertices α and $\alpha + 1$. By Lemma 4.1, unary has rank 1 for the gap α . □

Corollary 4.3 (failure of multi-gap pointwise optimality). *For any gap set A with $|A| = \ell \geq 3$, unary is not pointwise optimal: there exists a simplex binarization $B \in \mathcal{S}_d$ such that*

$$\text{rk}_B(A) = \lceil \log_2(\ell + 1) \rceil < \ell = \text{rk}_{B_U(d)}(A).$$

Example 4.4 (a smallest explicit counterexample). Take $d = 3$ and $A = \{0, 1, 2\}$. Define

$$B = \text{conv}\{(0, 000), (1, 010), (2, 100), (3, 111)\}.$$

The four binary vectors 000, 010, 100, 111 are affinely independent. Hence $B \in \mathcal{S}_3$. Projecting onto the first two coordinates gives

$$00, \quad 01, \quad 10, \quad 11,$$

which distinguishes all four blocks cut out by A . Therefore

$$\text{rk}_B(\{0, 1, 2\}) = 2.$$

By Lemma 4.1, however,

$$\text{rk}_{B_U(3)}(\{0, 1, 2\}) = 3.$$

5 Pareto-optimality for all single gaps

Although unary is not pointwise optimal for multi-gap rank, it is forced by the requirement of being optimal for every single gap simultaneously.

Proposition 5.1 (uniqueness under all single-gap optima). Let $B \in \mathcal{S}_d$ be represented by codewords $b^0, \dots, b^d \in \{0, 1\}^d$. Suppose that

$$\text{rk}_B(\{\alpha\}) = 1 \quad \text{for every } \alpha = 0, \dots, d-1.$$

Then, after permuting coordinates and possibly complementing individual coordinates, the code $b^0, \dots, b^d$ is the unary code.

Proof. Fix a gap α . The condition $\text{rk}_B(\{\alpha\}) = 1$ means that some single coordinate q hits every edge crossing the cut $\{0, \dots, \alpha\} \mid \{\alpha + 1, \dots, d\}$. Therefore

$$b_q^i \neq b_q^j$$

for every $i \leq \alpha < j$. Since this coordinate is binary, all vertices on one side of the cut have one common value and all vertices on the other side have the opposite value. In other words, coordinate q is a step function with its unique jump at the cut α .

A single binary coordinate cannot be a step function for two distinct cuts. Indeed, if it jumped at $\alpha < \beta$, then the entries in the middle region $\alpha + 1, \dots, \beta$ would have to be simultaneously equal to the right-side value for the first cut and the left-side value for the second cut, while the entries to the right of β would force those two values to be equal, a contradiction.

Thus the d cuts require d distinct coordinates, one for each cut. After permuting coordinates, coordinate q realizes the cut $q - 1$. Complementing coordinate q if necessary makes it equal to 0 on $\{0, \dots, q - 1\}$ and 1 on $\{q, \dots, d\}$. Therefore coordinate q of b^i is 1 exactly when $i \geq q$, which is precisely the unary step code. $\square$

Remark 5.2. Theorem 3.1 is a fixed- A statement: the minimizing simplex binarization may depend on the chosen set of gaps. Proposition 5.1 explains why no binarization can strictly dominate unary while also matching unary on every single-gap rank value.

6 Conclusion

The single-gap open problem has an affirmative answer: among natural simplex-isomorphic binarizations of $x \in \{0, \dots, d\}$, no binarization can have rank below 1 for a genuine gap, and unary attains rank 1 for every gap. For a fixed set of ℓ gaps, however, the exact optimum is $\lceil \log_2(\ell + 1) \rceil$, not ℓ . Thus the unary binarization is Pareto-optimal across the family of all single-gap ranks, but it is not pointwise optimal for the stronger multi-gap rank function.

References

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